Nidhi Jadhav

973-980-9103 | nidhivjadhav@gmail.com | linkedin.com/in/nidhii-jadhav | nidhijadhav.github.io

Education

Northeastern University Boston, MA
Bachelor of Science in Computer Science, Magna Cum Laude | ***GPA:** 3.83/4.0* *May 2026*
Concentration: Artificial Intelligence | **Minors**: Mathematics, Business
Awards: 50% Merit Scholarship, John Martinson Honors Program, Dean's List, PEAK Basecamp Research Award, PEAK Ascent Research Award, New Jersey Governor's School of Engineering and Technology

Experience

Software Engineer Intern | **Snowflake** Sept 2024 – Dec 2024
Machine Learning, LLM, RAG, Python, SQL, Streamlit, Snowflake, Solution Design
- Architected end-to-end ML pipeline for Snowflake Certified Solution analyzing 1,000+ sales transcripts for six enterprise customers using RAG-based LLM with vector embeddings and sentiment analysis
- Engineered Streamlit dashboard with natural language query interface, converting user prompts into optimized SQL and Python for real-time data visualization and analytics

Independent Researcher | **Goodwill Computing Lab** Sept 2025 – Present
Python, Data Analysis, Environmental Impact, Socio-economic Research, Datacenter Infrastructure
- Investigating intersection of datacenter infrastructure and socio-economic inequality by correlating intergenerational mobility metrics with economic and environmental impact datasets across US geographies
- Analyzing the geographic and demographic implications of data center placement on local communities and environmental resources

Student Researcher | **SimBioSys Lab** Jan 2023 – June 2024
Python, Flask, React, MySQL, Git, Research and Development
- Contributed to full-stack web application modeling the glycan shield enveloping viruses
- Developed PDB file parser to extract and process structural data on viral proteins, automating data pipeline for biomedical research workflows

Projects

Candor | *Python, HuggingFace PEFT, LoRA/QLoRA, FastAPI, pgvector, GitHub API* In progress
- Building a code review automation system that fine-tunes a local LLM on a repo's own PR history to post inline review comments on pull requests; targeting teams that want codebase-aware reviews without routing every PR through a cloud API
- Fine-tuned a code-specialized LLM using QLoRA via HuggingFace PEFT and TRL SFTTrainer on curated PR history; currently building the orchestration layer handling diff parsing, context assembly, and RAG retrieval of similar past review comments

Execution Decision Layer | *Next.js, TypeScript, Tailwind CSS, Anthropic API, Vercel* April 2026
- Designed and built a prototype agentic reliability system that routes proposed AI actions through a structured decision pipeline before execution, separating stochastic LLM signal estimation from deterministic policy application
- Implemented five-signal architecture (reversibility, ambiguity, external party, pending conditions, known exceptions) with graceful failure modes defaulting to confirmation to prevent irreversible actions under uncertainty; deployed on Vercel

ExamEngine | *Python, TypeScript, Next.js, FastAPI, PostgreSQL, AWS, Docker* Dec 2025
- Built full-stack exam scheduling platform for Northeastern's Registrar Office with Next.js/TypeScript frontend and FastAPI backend deployed on AWS; implemented DSATUR graph coloring algorithm with soft constraint optimization for conflict-free scheduling
- Achieved zero conflicts for 15,000+ students across 1,500+ course sections with sub-2-minute runtime, eliminating approximately 100 hours of manual scheduling per semester

Song Topic Classification from Lyrics | *Python, HuggingFace, RoBERTa, PyTorch, Scikit-learn* April 2026
- Fine-tuned RoBERTa (roberta-base) for 8-class song topic classification on 27,000+ songs, achieving 92% accuracy and 0.92 macro F1, outperforming TF-IDF + LinearSVC (90%) and Multinomial Naive Bayes (74%) baselines
- Analyzed per-class failure modes across imbalanced categories, identifying that model complexity matters most for minority and semantically overlapping classes such as feelings (F1: 0 to 0.88 from Naive Bayes to RoBERTa)

Emotion-Based Style Transfer on Visual Art | *Python, PyTorch, VGG-16, Neural Style Transfer* June 2021
- Trained CNN on ArtEmis dataset for emotion-conditioned style transfer, contributing novel approach to affective computing
- Authored research paper published in IEEE Xplore and presented at MIT URTC

Technical Skills

Languages: Python, R, Java, SQL, JavaScript/TypeScript, HTML/CSS, Swift
ML & AI: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, LLM fine-tuning, RAG, vector embeddings, Bayesian inference, MCMC, pandas, NumPy
Frameworks & Tools: Next.js, FastAPI, React, Node.js, Flask, Streamlit, Docker, AWS, CI/CD, GitHub Actions, Claude Code, PostgreSQL, MongoDB, Git